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REVIEW

Grime Review: What can remote sensing do for plant ecology?

Thermal remote sensing for plant ecology from leaf to globe

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Abstract

1. Surface temperatures are mechanistically linked to vegetation biophysical and physiological processes. Although remote sensing in the thermal infrared (TIR) domain can offer novel insights into the impacts of changing surface temperatures on vegetation, the transformative potential of remote sensing for plant ecology has not yet been realized.
2. Remotely sensed surface temperatures can be used to derive stomatal behaviour and identify stressful environmental conditions in near-real time. Plant species, traits and structural characteristics can be evaluated with high spectral resolution TIR emissivity.
3. Beyond canopy scales, thermal remote sensing can enhance the inferences obtained from manipulative experiments and empirical evidence, providing unique insight into shifts in species ranges and phenology with changing climate conditions.
4. Scaling leaf traits, canopy structure and regional patterns require an integrated understanding of both process and technology. Theory linking surface temperatures to vegetation dynamics is summarized from an energy balance perspective. We outline scaling considerations including the impacts of morphology on leaf energy balance, canopy structure influences on convective heat exchange and potential confounding impacts of non-vegetated surfaces.
5. *Synthesis*. We introduce a unifying framework to link leaf to globe through thermal remote sensing. Recent and emerging advances in sensors, data availability and analytics, together with synergies between TIR remote sensing and other data sources, present a timely opportunity for ecologists to advance our understanding of plant physiology, ecology and biogeography with thermal remote sensing.

KEYWORDS
climate change, plant stress, remote sensing, scaling, temperature, thermal infrared, vegetation

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1 | INTRODUCTION

Temperature is a fundamental component of climate and a key driver of vegetation distribution and functionality. Chemical and biological reactions are regulated by temperature, which influences all levels of organization in ecological systems. At the cellular level, high temperature extremes can cause irreversible damage to photosynthetic apparatus (Berry & Bjorkman, 1980; Teskey et al., 2015). Temperature, along with other biotic and abiotic factors such as nutrient and water availability, regulates biogeochemical cycles at ecosystem scales and influences the biogeographical distribution of plant and animal species at regional and global scales (Figure 1; Jeffree & Jeffree, 1994). Furthermore, anthropogenic climate change is expected to alter temperature regimes, with impacts on plants from leaf to global scales. Despite the importance of temperature across scales, evaluating trends across space and time is difficult. This is not a unique issue—scaling challenges are persistent and fundamental in all fields of ecology (Wiens, 1989). How do we understand the influence of photosynthetic kinetics on global vegetation patterns, or assess the impacts of periodic temperature extremes on future species ranges? One way to address scaling issues is to explore phenomena concurrently at multiple spatial and temporal scales. Thermal remote sensing is a key tool that can be used to investigate the relationships between temperature and vegetation at multiple levels of organization. Furthermore, thermal remote sensing is more accessible to plant ecologists than ever before—new sensors, increased data availability and emerging technologies could facilitate novel, cross-scale approaches to advance our understanding of temperature impacts on vegetation in the terrestrial biosphere.

Thermal remote sensing data can be obtained at various spatial, temporal and spectral resolutions coming from a range of platforms including satellites, aircraft, uncrewed aerial vehicles (UAVs), mounted on towers and ground-based systems including handheld cameras (Figure 2). Most thermal infrared (TIR) sensors used for vegetation analyses measure emitted radiation in the mid-range IR (3–8 μm) and, more commonly, the long-range IR (8–14 μm) regions (Kuenzer & Dech, 2013; Zhu et al., 2018). In remote sensing, there are inherent trade-offs among spatial, temporal and spectral resolutions of thermal instruments that must be considered when assessing a given science question. For example, UAV-based sensors can address questions at the leaf and canopy scale with single thermal pixels covering 40 cm or less (Figure 2), allowing differentiation between individual plants or even individual leaves. Conversely, spaceborne sensors provide global thermal coverage of the earth's surface, but the spatial resolution of pixels has been typically greater than 60 m, limiting application to canopy scales and larger (Kuenzer et al., 2013). Spaceborne satellites in geostationary orbits can obtain thermal imagery every 15 min, but the spatial resolution is typically too coarse for canopy scale measurements (>2 km); higher spatial resolution sensors typically sacrifice high temporal resolution (Fisher et al., 2017). Clouds are problematic for both air- and space-borne sensors as they can scatter, absorb and re-emit long and short-wave radiation, confounding temperature estimates

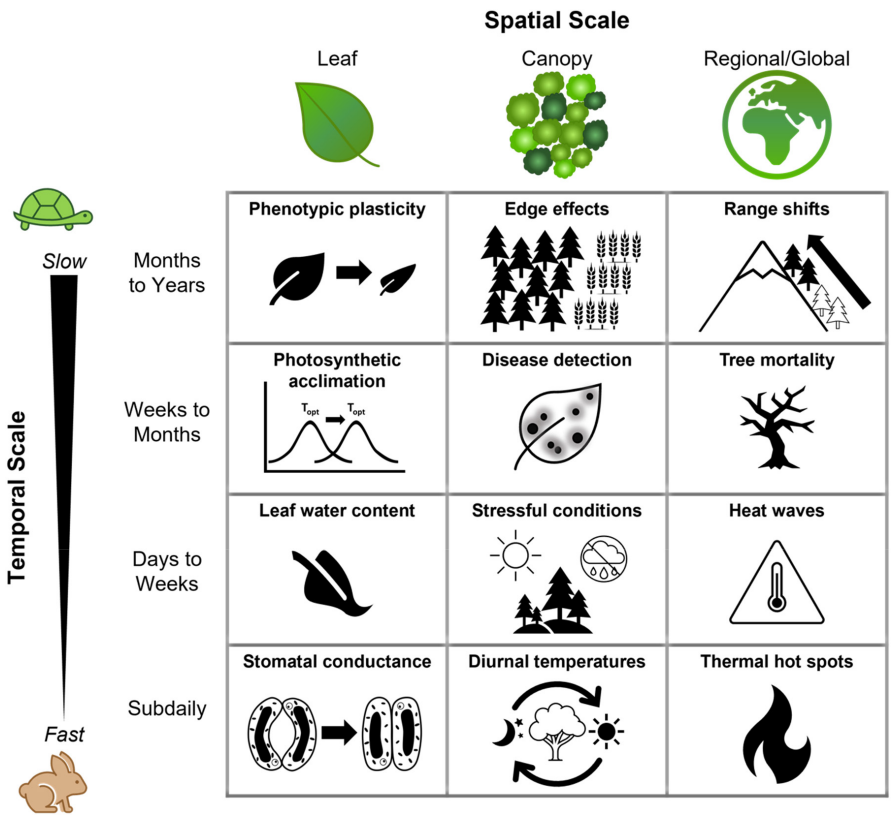

FIGURE 1 Opportunities for thermal remote sensing to inform plant ecology across spatial and temporal scales.

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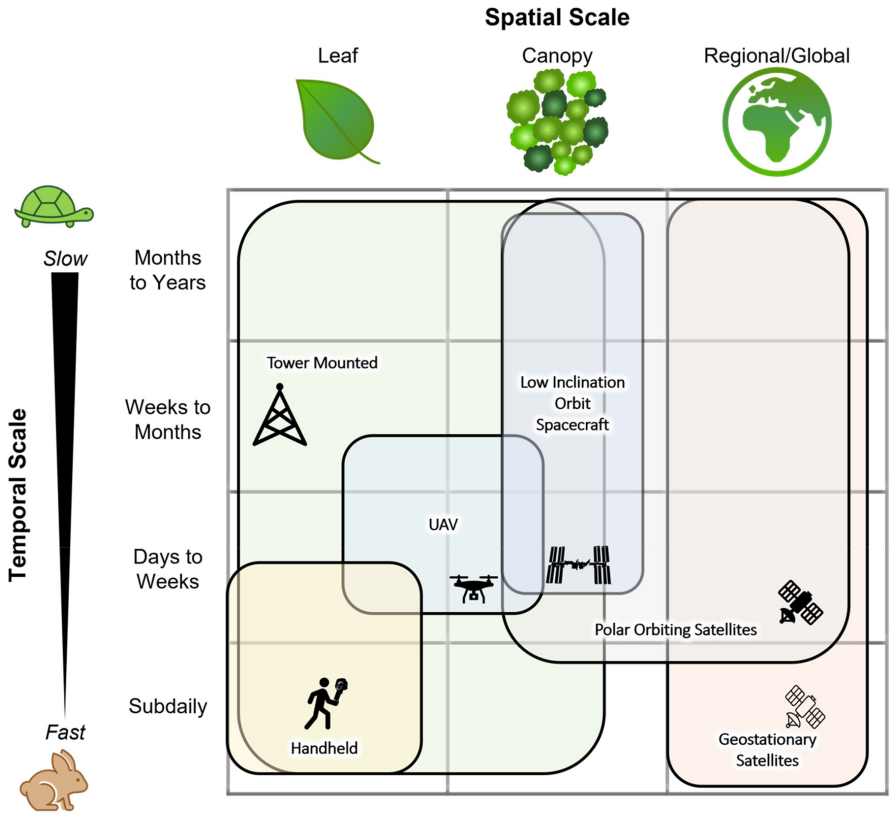

FIGURE 2 Spatial and temporal coverage of thermal infrared remote sensing methods for obtaining estimates of surface temperature.

obtained from thermal retrievals. With low-altitude UAVs, clouds are less of an issue but there are logistical challenges associated with dedicated long-term sampling to observe slowly changing ecological phenomena at canopy scales (Figure 2). Despite the trade-offs, with careful consideration of confounding factors, scaling issues and the selection of appropriate data sources, thermal remote sensing data can help address important questions in plant biology across spatial scales from leaf to globe (Figure 1).

One of the most ecologically useful products derived from thermal remote sensing data is surface temperature (T_{surf}) estimates. Surface temperatures both impact and respond to a variety of chemical, biological and physical properties, providing valuable insight into plant patterns and underlying plant controls in ecosystems (Still, Powell, et al., 2019). For example, when plants experience stressful heat and moisture conditions, they respond by closing stomata, which causes reduced transpiration and associated increases in leaf T_{surf}. Remote sensing of T_{surf} can help identify plants experiencing heat and water stress (Buitrago et al., 2016; Coates et al., 2015). Across larger scales, thermal data can improve plant land cover classifications and can be leveraged to evaluate spatial distributions of certain species, communities or management regimes (Barnes et al., 2021). Overall, remote sensing of T_{surf} provides valuable information into the water-energy exchange between plants, soil and the atmosphere.

The thermal radiation emitted from an object depends on both its temperature and emissivity; as such, emissivity is required to retrieve T_{surf} from TIR measurements. Emissivity is the effectiveness of a material to emit and absorb thermal radiation relative to a blackbody, and is generally close to 1 (0.96–0.99) for healthy vegetation, but can be as low as 0.88 to 0.94 for dry leaves (Kuenzer & Dech, 2013). Accurate T_{surf} measurements from thermal remote sensing require calibrations that account for the impact of emissivity on radiometric temperatures (Fuchs & Tanner, 1966).

Surface temperatures can deviate substantially from air temperatures (T_{air}), yet T_{air} are commonly used to describe plant temperature in plant ecological studies (Körner & Hiltbrunner, 2018). Near-surface T_{air} (2 m above the land surface) are used to assess vegetation–climate interactions at regional and global scales in models, projections and gridded climatologies. Gridded T_{air} estimates typically rely on data from weather stations, with uneven spatial coverage (Menne et al., 2012). In contrast, spaceborne TIR measurements, used to compute T_{surf} (also called land surface temperature; LST), are available at high spatial resolutions and global coverage (Duan et al., 2017; Lawrimore et al., 2011). Plant biological and biophysical properties can influence decoupling between T_{air} and T_{surf}. In densely vegetated forests, deviations between T_{air} and T_{surf} (ΔT) are generally small compared to ΔT differences observed in grasslands, savannas, heath and tundra systems with smaller stature vegetation (Mildrexler et al., 2011). As such, T_{surf} estimates more accurately reflect the extreme and dynamic thermal environment experienced by plants and other organisms living on the land surface. Furthermore, T_{surf} provides insight into facets of plant physiology including growth, stomatal conductance, stress and transpiration. Despite its more direct relevance to plant biological processes,

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regional and global-scale studies of terrestrial vegetation seldom examine T_{surf} explicitly, presenting an exciting opportunity for the use of thermal remote sensing for studies of plant physiology, ecology and biogeography at leaf, canopy and broader scales.

Here, we consider three ecological scales (Figure 1) where thermal remote sensing has strong potential to advance understanding of plant physiology and ecology. Each level of organization—leaf, canopy and beyond canopy (e.g. regional, global) scale—has its own theoretical underpinnings, methodological and scaling considerations, and scientifically motivating future directions. We also highlight specific opportunities where thermal remote sensing could address open questions in plant ecology. We conclude by delving into principles of thermal remote sensing and outline current and future technologies that could be most useful to ecologists. We hope this review will serve as both a bridge between thermal remote sensing and plant ecology and as motivation for ecologists to consider incorporating thermal remote sensing into studies of plant dynamics.

2 | THERMAL REMOTE SENSING FOR PLANT ECOLOGY ACROSS SPATIAL AND TEMPORAL SCALES

2.1 | Leaf scale

2.1.1 | Background

The leaf energy budget governs leaf temperature, which changes in response to biotic and abiotic factors. Leaf temperature (T_{leaf}) is determined by the balance between incoming energy to a leaf (shortwave and longwave radiation) and outgoing energy due to sensible and latent heat fluxes, and reflected and emitted radiation (Jones, 2013a, 2013b; Lambers & Oliveira, 2019; Nobel, 2009). During the growing season, T_{leaf} often deviates substantially from T_{air} in the short term (i.e. diurnally; Leuzinger & Körner, 2007; Smith & Carter, 1988), with temporally persistent deviations between T_{air} and T_{leaf} also reported (Michaletz et al., 2016).

Stomatal movements influence leaf water loss and thus the leaf energy budget (Jones, 2013a, 2013b), but stomatal responses to dynamic natural environments are complex and not fully understood (Buckley & Mott, 2013). Stomatal closure, which can be a plant response to stressful abiotic conditions including water deficits caused by low soil moisture (Katul et al., 2012) and/or high atmospheric vapour pressure deficit (Oren et al., 1999), affects the difference between T_{air} and T_{leaf}. High transpiration rates can cool the leaf surface relative to ambient conditions, which can result in T_{leaf} that is cooler than T_{air} (Cook et al., 1964; Gates, 1968). Conversely, stomatal closure results in a reduction of transpiration, modifying the leaf energy budget such that T_{leaf} can be hotter than T_{air} (Leigh et al., 2017). Leaf temperature therefore provides direct insight into plant ecophysiological processes including rates of photosynthetic carbon dioxide assimilation and transpiration (Dai et al., 2004; Gates, 1968; Kobza & Edwards, 1987; Scafaro et al., 2021; Yang et al., 1990). High accuracy and precision thermal remote sensing are therefore necessary to differentiate these T_{air} and T_{leaf} differences.

2.1.2 | Scaling considerations

Many structural factors beyond stomatal conductance influence T_{leaf}, which can complicate direct estimates of stomatal activity from leaf-level thermal imagery (also called thermography). Leaf angle and canopy position, for example, can influence how much direct solar radiation a leaf receives, with leaves lower in the canopy having more shade and less direct sunlight even though they might have larger leaf areas and flatter angles relative to the sun (Harley et al., 1995; Medina et al., 1978). Leaf trichomes or 'hairs', which occur across taxa with high morphological variation (Johnson, 1975), can impact the leaf energy budget and thus T_{leaf} through impacts on both the leaf boundary layer and leaf optical properties (Ehleringer et al., 1976; Schuepp, 1993). In xeric environments, densely pubescent leaves can increase light reflectance across wavelengths (Ehleringer et al., 1976) resulting in lower leaf temperatures (Ehleringer & Mooney, 1978; Skelton et al., 2012). Conversely, increased boundary layer thickness from leaf pubescence can lead to increased T_{leaf} through increased boundary layer resistance to convective and latent heat transfer (Meinzer & Goldstein, 1985; Ripley et al., 1999).

Leaf size also influences the energy budget. Plants in xeric environments have evolved small leaves, compared to plants from systems that receive more annual rainfall, which allow for rapid convective heat exchange and reductions in T_{leaf} when T_{air} is high (Peguero-Pina et al., 2020; Yates et al., 2010). Temperatures can vary significantly within a single leaf with a temperature differences of up to 14.7°C observed across a single Proteaceae leaf (Leigh et al., 2017). Temperature differences within leaves depend on leaf size, with larger leaves having a higher range of temperatures than smaller leaves due to more uneven distribution of water across the leaf, higher surface complexity leading to a larger range of sun angles and a greater variation in boundary layer resistance that increases with distance from the leaf centre and negatively influences convective heat exchange (Leigh et al., 2017; Raschke, 1960). Accurate consideration of leaf structural traits can improve scaling of leaf energy budgets to canopy and beyond canopy-scale temperatures, thus allowing thermal remote sensing to provide unique insight into open questions in plant biology at the leaf level.

2.1.3 | Thermal remote sensing: Opportunities

Opportunity 1: Non-uniform stomatal responses

Estimating leaf stomatal conductance from thermal imagery could resolve measurement-speed bottlenecks inherent to traditional in situ leaf gas exchange methods (Araus & Cairns, 2014; Prado et al., 2018;

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Rischbeck et al., 2017) by rapidly capturing many leaves in a single field of view, allowing for the analysis of spatial and temporal trends in stomatal conductance (Jones, 1999). Leaf to air temperature differences ($T_{leaf} - T_{air}$) obtained from thermal imagery have been widely used as indicators of stomatal conductance and plant water status, particularly in agricultural systems (Fuchs, 1990; Inoue et al., 1990; Jones, 1999; Leinonen et al., 2006; Lima et al., 2016). Widely used Penman–Monteith-based (Monteith, 1965; Penman, 1948) approaches to estimating stomatal conductance from leaf thermography assume steady-state environmental conditions (Monteith & Unsworth, 2013).

Methods to estimate stomatal conductance and water status from leaf-level thermal imagery could yield insights into the impacts of non-uniform stomatal responses on photosynthetic rates. Vialet-Chabrand and Lawson (2019) circumvented steady-state assumptions by incorporating a suite of differential equations to derive stomatal conductance from leaf thermography under varying environmental conditions. Stomata on a leaf can respond heterogeneously in space (Downton et al., 1988; Mott et al., 1993; Terashima, 1992) and time (Matthews et al., 2017; Vialet-Chabrand et al., 2017; Vialet-Chabrand & Lawson, 2020), often resulting in 'stomatal patchiness'. The impacts of stomatal patchiness on photosynthetic rates (Matthews et al., 2017) and how non-uniform responses scale up to impact whole-plant photosynthesis is unresolved. Estimates of stomatal conductance from leaf thermography could allow for evaluation of variation in stomatal patchiness across multiple plants, species and biomes. Incorporating leaf-level thermal imagery in the field could relate spatial and temporal variation in stomatal conductance with environmental variation and plant water stress. An additional line of inquiry could examine the validity of current model assumptions of instantaneous stomatal response across a variety of species (Damour et al., 2010). We might expect that spatial averaging of carbon assimilation that does not consider spatial and temporal dynamics of stomatal responses might overestimate modelled photosynthesis in situations where stomatal conductance strongly limits carbon uptake.

Opportunity 2: Systematic deviation between T_{leaf} and T_{air}

Whether plants have the capacity to regulate T_{leaf} is a controversial topic in plant biology. The terms 'thermoregulation' and 'homeothermy' are widely used to describe the idea of regulating T_{leaf} relative to T_{air} through intrinsic mechanisms; however, we refrain from applying these terms here because both terms have established physiological meaning that conflicts with their colloquial use in plant biology. Interspecific differences in systematic deviation between T_{leaf} and T_{air} may be caused by differences in functional and structural leaf traits that influence the leaf energy balance (Fauset et al., 2018). Some studies have found evidence for systematic deviation between T_{leaf} and T_{air}, where T_{leaf} is cooler than T_{air} at high temperatures, and T_{leaf} is warmer than T_{air} at low temperatures (Blonder & Michaletz, 2018; Dong et al., 2017; Michaletz et al., 2015, 2016). Such systematic deviations between T_{leaf} and T_{air} are also consistent with leaf energy budget models that assume no change in stomatal opening (Jones, 2013a, 2013b), and thus are not evidence for intrinsic regulation of T_{leaf} on their own. There are few studies on temporally persistent deviations between T_{leaf} and T_{air} outside of controlled environments like leaf chambers (Blonder & Michaletz, 2018; Drake et al., 2020; Still, Sibley, et al., 2019). It is unknown whether such deviations extend beyond the scale of individual leaves (Heidkamp et al., 2020). A recent whole-tree chamber experiment found no evidence for T_{leaf} regulation in eucalyptus; T_{air} strongly controlled T_{leaf}, with deviations driven by the environment (Drake et al., 2020). Furthermore, evidence for intrinsically driven persistent differences between T_{leaf} and $T_{air.}$ based on leaf gas exchange measurements was recently called into question due to artefacts induced by gas exchange system design that produce results consistent with leaf homeothermy even when no leaf is in the chamber (Still, Sibley, et al., 2019). Thermal remote sensing measurements of T_{leaf} from handheld thermal imagers or UAVs could help resolve controversy surrounding T_{leaf} regulation. Experiments that relate T_{leaf} to T_{air} across a range of species and biomes could be used to develop a remote-sensing-based index of leaf thermal responses. The slope of the relationship between T_{leaf} and T_{air} (Blonder & Michaletz, 2018) in varying thermal environments could be particularly useful for assessing plant ecological strategies.

2.2 | Canopy scale

2.2.1 | Background

Integrating across leaves, thermal data facilitate the upscaling of modelled and observed biophysical processes from individual leaves to ecosystems. Canopy temperature (T_{can}) measurements, which can be obtained with thermal imaging, represents the aggregate temperature of multiple leaf assemblages that are directly in a sensor's field of view (Still et al., 2021). Canopy temperatures change throughout the day as the balance shifts between incoming and outgoing longwave and shortwave radiation to and from the canopy, evapotranspiration (ET) and sensible heat exchange (Hatfield et al., 1984; Huang, Cai, et al., 2019; Peterschmitt & Perrier, 1991). Relative to the leaf level, there are far fewer studies that have evaluated remotely sensed emitted radiation at the canopy scale due to attenuation of TIR spectra at this scale (Ribeiro da Luz & Crowley, 2007). The most common application of remotely sensed T_{can} estimates has been to assess plant stress and guide management practices in agricultural systems (Gerhards et al., 2019). With advances in sensor resolutions and data availability, there will be more opportunities to extend leaf-level studies for early identification of plant stress in natural systems.

2.2.2 | Scaling considerations

Multiple factors simultaneously influence T_{can}, including abiotic factors (e.g. solar radiation, vapour pressure deficit, T_{air}, wind speed),

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canopy structure and thermal radiation from surrounding and underlying objects (Kim et al., 2016). The relative proportion of sunlit and shaded leaves, for example, can strongly influence T_{can} (Meron et al., 2013). Yi et al. (2020) recently found that T_{can} values within a single tree are impacted by leaf morphology and foliage clumping. More homogeneous T_{can} values were observed in a pine species relative to broader-leaf oak species, which was attributed to higher transmittance of solar radiation throughout the canopy (Yi et al., 2020). Tree density influences sensible and latent heat fluxes, where denser canopies support higher fluxes and larger T_{can} to T_{air} differences than sparser canopies (Aubrecht et al., 2016; Leuzinger & Körner, 2007; Mildrexler et al., 2011). Large differences between T_{can} and T_{air} have also been observed in short stature vegetation and homogeneous canopies where air under the top of the canopy becomes decoupled from air above the canopy, lowering the rates of convective heat exchange (Leuzinger & Körner, 2007; Scherrer & Körner, 2009). The surrounding environment also impacts T_{can} values with higher daily mean T_{can} to T_{air} differences observed from trees that were surrounded by higher fractions of impervious surfaces (Meier & Scherer, 2012). Linking remotely sensed TIR measurements to large-scale plant ecophysiological processes requires information on canopy structure and how the surrounding environment influences T_{can} across space and time.

Underlying objects such as soil and understorey vegetation play important and often overlooked roles in T_{can} variability. A positive relationship between T_{can} and soil temperature has been observed, with weaker correlations during sunny periods attributed to larger T_{can} changes and smaller soil temperature changes with low radiative heating of soils under dense canopies (Kim et al., 2016). Soil temperature is especially important in systems with sparse vegetation like drylands, where bare soil might make up a significant portion of the total energy emitted from a mixed pixel, or where a significant portion the soil heat flux reaches the canopy through short vegetation (Ogée et al., 2001; Smith et al., 2019). Vegetation in the bottom of a canopy can have higher temperatures than top-of-canopy vegetation when soils and other underlying vegetation are hot, dry and radiating large amounts of heat. In these circumstances, T_{can} may not accurately represent bulk T_{can} when measured from above by aircraft or satellites (Johnston et al., 2021). It is likely that many individual properties of the soil and understorey vegetation can simultaneously impact top-of-canopy T_{can} estimates measured with remote sensing. The relative importance of these factors and their contribution to T_{can} estimates depend on properties of the ecosystem evaluated such as vegetation cover and time of year.

2.2.3 | Thermal remote sensing: Opportunities

Opportunity 1: Monitoring of stressful conditions

Remotely sensed T_{surf} retrievals have been used to assess environmental conditions associated with plant stress across a variety of ecological systems. The mechanisms that link T_{leaf} to transpiration provide the basis for using T_{can} to identify areas of potential plant stress. When plants are exposed to unfavourable abiotic conditions such as high temperatures and/or water limitation, one potential response is stomatal closure to prevent excess water loss and subsequent development of plant stress (Jones, 2013a, 2013b). Low soil moisture and high atmospheric demand can trigger stomatal responses detectable within minutes of the onset of stressful environmental conditions, days or weeks before plants display visible symptoms (Lawson & Blatt, 2014; Santesteban et al., 2017; Zhang et al., 2019). Remotely sensed measurements of T_{can} have been widely used in crop studies to assess ET, monitor stressful conditions and guide irrigation interventions in agroecosystems (Anderson et al., 2011; Clawson et al., 1989; Jackson et al., 1981; Patel et al., 2001). The widely used Crop Water Stress Index (Idso et al., 1981; Jackson et al., 1981) leverages the difference between T_{can} and T_{air} relative to the differential between a well-watered crop and a non-transpiring canopy to assess crop stress. Another stress metric, the Evaporative Stress Index (ESI), can be based on TIR estimated ET and is an early indicator that captures the intensity and magnitude of drought stress experienced by plants (Anderson et al., 2016; Fisher et al., 2017; Otkin et al., 2013). Remotely sensed LST has been used to estimate ET as a function of the surface energy budget since the early 1970s (Bartholic et al., 1972). The history and evolution of these methodologies over time is beyond the scope of this review, but we direct interested readers to existing reviews on these concepts (Chen & Liu, 2020; Wang & Dickinson, 2012).

Despite the established use of thermal data to monitor environmental stress in agricultural systems, there is a paucity of studies that systematically leverage these data to evaluate stressful conditions in less-managed environments. Forest responses to changing climate conditions remain a large uncertainty in future climate projections (Bugmann et al., 2019; Krause et al., 2018; Kurz et al., 2008). Widespread tree mortality in response to drought, increasing temperatures and drought-induced susceptibility to pests and diseases has been well documented (Anderegg et al., 2012; Breshears et al., 2005; Phillips et al., 2009, 2010; Rosenzweig et al., 2001; Schuldt et al., 2020). With drought frequency and intensity expected to increase under changing climate conditions with ecosystems struggling to fully recover before the next drought (Pachauri et al., 2015; Schwalm et al., 2017), early detection of plant stress before widespread mortality occurs could have major scientific and societal benefits. Recently, Smigaj et al. (2019) leveraged thermal data to evaluate disease severity in a pine plantation. Using thermal data acquired with a UAV, they found significant relationships between a metric calculated from T_{can} measurements and red band needle blight severity in a plantation in Scotland (Smigaj et al., 2019). The use of thermal data to evaluate disease and stress at the canopy scale in natural systems is in its infancy. TIR remote sensing data could allow for early detection of stressed trees, enable in situ monitoring of trees experiencing stressful conditions, increase our understanding of tree mortality and ecological responses to climate extremes, and enable better monitoring and effective management interventions (Anderegg et al., 2019; Huang, Anderegg, & Asner, 2019).

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Opportunity 2: Forest edges

Thermal data also facilitate monitoring of variable T_{can} at different locations in a forest. Although 70% of the world's forests are <1 km from an edge, most of our understanding of forest dynamics comes from studies on intact forest systems (Haddad et al., 2015). Relative to forest interiors, forest edges have higher light availability, T_{air}, soil temperatures, vapour pressure deficit and wind (Figure 3; Chen et al., 1993; Denyer et al., 2006; Matlack, 1993). Canopy temperature of edges relative to interior forests is unresolved with both higher and lower T_{can} observed depending on surrounding objects and time of year (Pohlman et al., 2007). The changes in environmental conditions at forest edges can impact rates of photosynthesis and transpiration in these areas. In tropical systems, for example, carbon stocks of edges are lower than those of interior forests due to increased heat and drought stress caused by higher T_{air} in these areas (Ordway & Asner, 2020; Smith et al., 2018). Higher temperatures at edges may be beneficial in boreal systems, where edge effects could increase the number of days near optimal temperatures for photosynthesis (Harper & Macdonald, 2001; Smith et al., 2018). Conversely, increased atmospheric demand associated with higher temperatures could increase vulnerability to stress for trees at edges. Logging operations, increases in agricultural lands and the rising human population are expected to continue to increase the global amount of fragmented forests (Bryan-Brown et al., 2020; Harper et al., 2007; Kirby et al., 2006).

Thermal remote sensing could help us evaluate transpiration and stressful conditions at forest edges. In particular, UAV flights across transects that extend from forest interior across forest edges at multiple timepoints could help characterize differences in the thermal environment between edges and forest interior. Horizontal thermal profiles could be used to evaluate seasonal and weather-driven shifts in the temperature differential between forest edges and interiors.

Opportunity 3: TIR Emissivity

Beyond T_{can} estimates derived from emitted TIR radiation, emissivity over the entire TIR region can provide valuable insight into plant ecology. In contrast to TIR reflectance, which measures the amount of thermal energy reflected or emitted off a source, TIR emissivity measurements use the sample as the thermal source and compare emitted energy to a blackbody (emissivity = 1) radiating at the same temperature. Although spectral patterns in TIR reflectance and TIR emissivity are often similar, emissivity is generally more noisy and directly analogous to measurements made by airborne and spaceborne sensors (Byrnes & Byrnes, 2011). Ullah et al. (2012) found emissivity spectra from various wavebands in the mid-wave infrared (MWIR; 2.5–6 μm) and longwave infrared (LWIR; 8–14 μm) regions to be statistically different among 13 common broadleaf species and used these differences to identify individual species. In the field, Rock et al. (2016) found that, at high signal-to-noise ratios, hyperspectral TIR data had higher plant classification accuracy than visible-shortwave infrared (VSWIR; 380–2500 nm) data. Meerdink et al. (2019), however, recently showed that only 11 out of 24 species evaluated were spectrally distinguishable, suggesting that TIR emissivity may not always be able to be used to differentiate individual species. Differences in TIR emissivity among species are likely due to variation in biophysical and biochemical constituents of leaves. Changes in lignin, cellulose, leaf water content and leaf area can all generate measurable changes in TIR spectra (Buitrago Acevedo et al., 2017). Recently, however, Richardson et al. (2021) showed that changes in leaf cuticular development related to leaf age and development stage, rather than internal structural or biochemical constituents, were the main drivers of differences in TIR emissivity for five temperate deciduous tree species. Although in its infancy, the use of TIR emissivity to evaluate individual plant species or traits illustrates the importance of spectral resolution.

TIR emissivity signals at the canopy scale can be complicated by structural and soil factors that require careful consideration. Thermal emissivity from the canopy is higher than emissivity from any individual component of the plant due to the 'cavity effect' caused by radiative scattering and absorption in tree canopies. This is especially important to consider at intermediate vegetation covers to avoid underestimation of canopy emissivity (Jacob et al., 2017). A positive relationship between leaf area index (LAI) and TIR emissivity has been observed with possible species-specific saturation at high LAI values. Soil water content dramatically impacts emissivity with over an 800% increase in emissivity at higher water contents observed, especially in sandy soils (Mira et al., 2007). Even small amounts of atmospheric humidity absorbed by soils in early morning hours can impact emissivity estimates obtained from Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) data, illustrating the importance of soil moisture measurements for TIR validation experiments (Mushkin et al., 2020). Soil emissivity has a greater impact on canopy emissivity in systems where LAI and NDVI values are low (Olioso, 1995). Although it is well recognized that soil emissivity impacts canopy emissivity, there is a paucity of studies that have evaluated how variation in soil properties impacts airborne and spaceborne TIR data. Future studies that combine simultaneous measurements of soil physicochemical dynamics with canopy-level TIR data can help to resolve these outstanding questions and result in more accurate retrievals of canopy emissivity estimates.

2.3 | Beyond canopy scale

2.3.1 | Background

Thermal remote sensing can inform understanding of how plants respond to their environment at regional and global spatial scales. The impacts of warming-induced shifts in temperature regimes and phenology on the terrestrial biosphere are becoming more evident across spatial and temporal scales (Hatfield & Prueger, 2015; Körner, 2021). Temperature and water availability are the primary drivers of the geographical ranges and distribution of species (Box, 1981; Guisan & Zimmermann, 2000; Pearson & Dawson, 2003). At regional and global scales, rising temperatures can induce shifts in biogeographical distribution of plant species (Chen et al., 2011), generally forcing shifts

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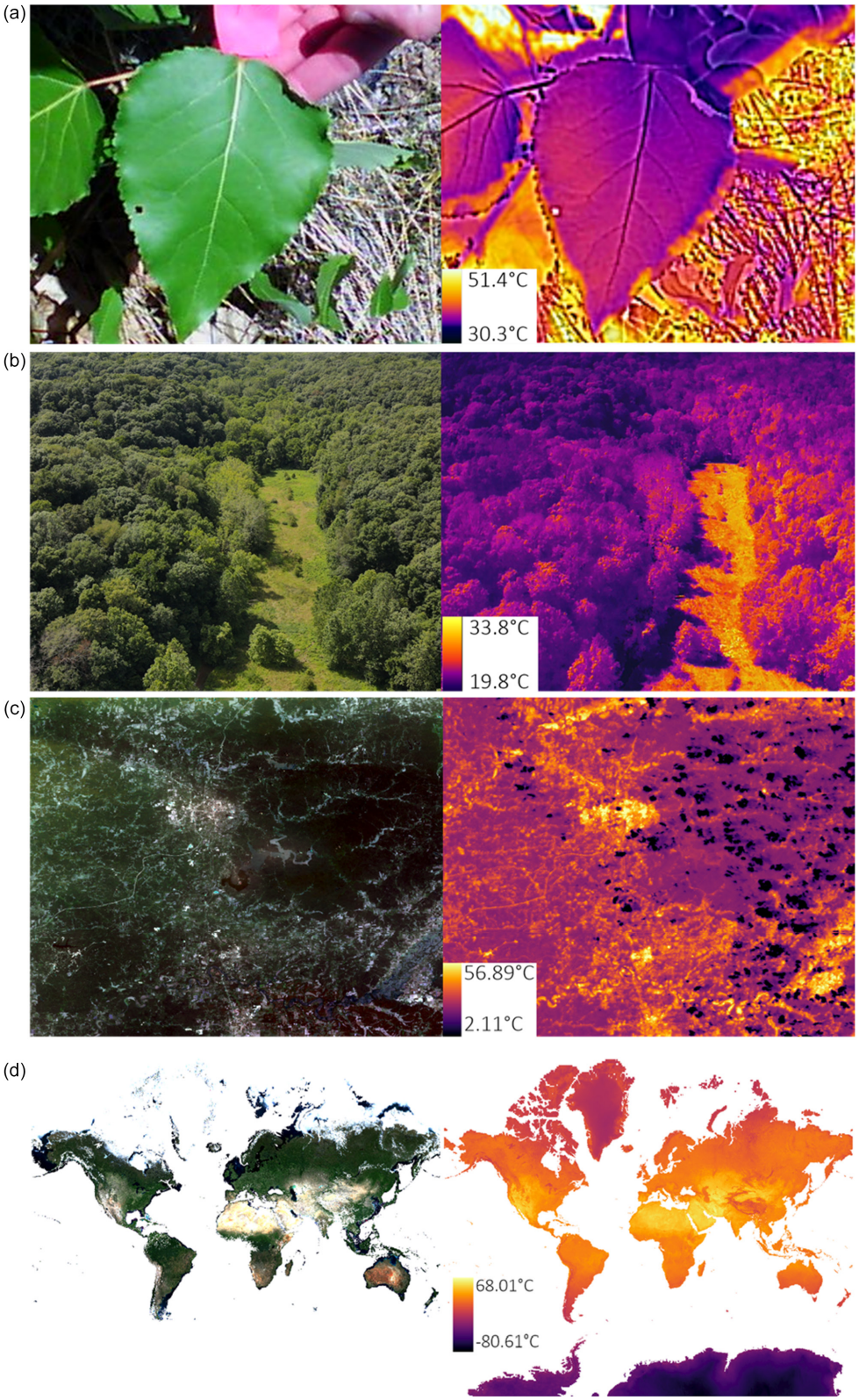

FIGURE 3 Thermal data from leaf to global scales (right column) alongside true colour images (left column). (a) Leaf-level thermal imagery acquired with a handheld FLIR thermal camera. (b) Canopy-level thermal and true colour drone imagery acquired with a DJI Mavic 2 Enterprise advanced UAV June 2021 outside of Bloomington, IN illustrating a forest: Grassland boundary. (c) Regional Landsat 8 RGB imagery from 24 June 2021 and thermal data from the ECOSTRESS daily LST 70 m data product for 16 June 2021. (d) Global RGB imagery from MODIS MCD43A4 500 m data product averaged over June 2021 and land surface temperature data from the MODIS MOD11C3 monthly 0.05° LST data product for June 2021.

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poleward or upwards in elevation (Anderson & Wadgymar, 2020; Lenoir & Svenning, 2015) as plants become locally maladapted to their former ranges (Wilczek et al., 2014). Although the fossil record has shown positive relationships between biodiversity and temperature (Mayhew et al., 2012), present-day increases in global temperatures resulting from anthropogenic climate change threaten current levels of biodiversity and ecosystem functionality due in part to shifts in phenology, species distributions and new trophic networks that are disadvantageous (e.g. lack of pollinators or introduction of novel defoliators) (Bellard et al., 2012; Nunez et al., 2019).

2.3.2 | Scaling considerations

The impacts of rising temperatures on vegetation are highly variable across the planet. Variation in abiotic and biotic factors across space and time adds further complexity to plant species responses to climate change. For example, the economically and ecologically important silver fir (*Abies alba* Mill.) may thrive under predicted climate warming in Western and Central Europe due to its tolerance to warm temperatures and high resilience and resistance to drought conditions relative to its competitors (Vitasse et al., 2019). Conversely, large tropical alpine plants like *Lobelia rynchopetalum* face very high extinction risk with climate warming due massive warming-induced range reductions and associated losses of genetic diversity (Chala et al., 2016). It is difficult to apply or scale a value observed at an instantaneous point of time to longer timeframes (Chen et al., 1999; Jarvis, 1995). This is especially problematic given the large degree of heterogeneity in the speed and degree of warming across the planet that is expected (Ji et al., 2014). Inconsistency in observations and sampling across various biomes and plant communities also present challenges. For example, over 90% of the observations in the TRY global plant database come from temperate and tropical systems, making inferences to other systems difficult (Kattge et al., 2011). Together, these challenges make it difficult to apply results from a single system, species, biome or study and predict climate impacts across the entire terrestrial biosphere. Thermal remote sensing, however, presents an opportunity to overcome geographical, logistical and financial constraints on experiments by observing the terrestrial biosphere continuously in time and space.

Manipulative warming experiments are crucial to investigating responses of leaves and canopies to climate change (Bagley et al., 2015; Liang et al., 2013). Warming studies typically manipulate temperatures in growth chambers (Gaihre et al., 2013; Grogan & Chapin III, 2000; Gunderson et al., 2000), greenhouses (Henry & Molau, 1997; Sistla et al., 2013) and field plots (Gunderson et al., 2010; Kimball et al., 2018; Niu et al., 2008; Reich et al., 2018) to study specific plant species, communities or ecosystems. The specificity and short duration of many warming experiments make it difficult to scale up results from individual warming experiments to inform large-scale terrestrial biosphere responses to climate change (Shiklomanov et al., 2019). Small-scale experiments are constrained in terms of time, space and the number of driver variables analysed or manipulated, and thus cannot account for the increasing number of feedbacks and interactions that occur at large spatial scales (Leuzinger et al., 2011). Due to high costs and labour requirements, many of these studies do not occur over the recommended duration of ≥10 years necessary to detect impacts of global change drivers on plant community structure (Komatsu et al., 2019). Such spatial and temporal scaling issues may lead to both overestimates (Leuzinger et al., 2011) and underestimates (Komatsu et al., 2019; Muller et al., 2020; Wolkovich et al., 2012) of the magnitude of vegetation response to climate change. Additionally, manipulations performed in experimental warming studies are generally performed on seedlings or saplings (Lu et al., 2013) due to the immense logistical and financial difficulty in conducting warming experiments in mature forests (Cavaleri et al., 2015).

Thermal remote sensing could enhance and extend the results of manipulative warming experiments. Incorporating canopy-level thermal data from UAVs would be a repeatable and relatively simple approach to assessing T_{surf} of multiple mature trees. Using thermal remote sensing data to empirically upscale coordinated warming experiments could facilitate analysis across sites spanning diverse plant communities and abiotic conditions. For example, T_{surf} from spaceborne remote sensing instruments could be used to extrapolate findings from temperature acclimation experiments in the context of natural intra- and inter-annual variability in T_{surf}. Furthermore, thermal data would allow for the evaluation of the impacts of long-term changing thermal regimes on plant communities.

2.3.3 | Thermal remote sensing: Opportunities

Opportunity 1: Range shifts

Vegetation may also undergo range shifts to track suitable climate spaces as global temperatures rise and climate variability increases (Germain & Lutz, 2020; Parmesan & Yohe, 2003). Species with high thermal acclimation potential appear to have a lower degree of observed range shift, reflecting reduced pressure to track their historical climate niches (Diamond, 2018). Increasing temperatures with climate change have already caused range shifts in some terrestrial plants with further poleward and upslope shifts anticipated (Gottfried et al., 2012; Parmesan & Yohe, 2003). There is a strong relationship between the rate of mean yearly temperature increase and rate of species distribution changes (Chen et al., 2011), with high variation within taxonomic groups that necessitates characterization of many individual species. Warming-induced shifts in vegetation distributions may be particularly apparent at ecotones—areas of transition between ecosystems (Risser, 1995). Climate change-induced shifts in plant species distributions have potential consequences for ecosystem functional properties including carbon and nutrient cycling (Gonzalez et al., 2010; Grimm et al., 2013). A recent biogeographical study found that the thermal environment at the leaf level and decoupling of T_{leaf} from T_{air} was a strong determinant of elevational distributions of plants (Liancourt et al., 2020), underscoring the importance of T_{surf} in regional scale studies of vegetation

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distributions. Thermal remote sensing can be used to characterize the thermal environment of a given species, ecosystem or biome in space and time. Additionally, the availability of 20+ years of continuous estimates of LST from satellites could be used to detect shifting thermal regimes through time. Areas with the most rapid shifts in thermal regimes could be flagged for vulnerability assessments. At ecotones and the edges of species ranges, the thermal environment from 20 years ago could be compared to the current thermal environment to assess the speed of vegetation shifts. Remotely sensed Tsurf, rather than air temperatures (e.g. O'sullivan et al., 2017), could yield more accurate estimates of thermal safety margins and thus better characterize the vulnerability of plant species to future warming. Long-term (25+ years) T_{surf} timeseries are available from spaceborne satellites at various spatial and temporal resolutions. Geostationary satellites, which obtain measurements of the same location many times in a 24-h period, have the necessary temporal resolution to assess low and high temperature extremes that drive species range shifts (Parmesan et al., 2000; Zimmermann et al., 2009). Yet, the coarse spatial resolution of geostationary satellites is detrimental to sensing extremes because low and high temperatures within the same pixel are blended. Estimates of the thermal range for species and ecosystems could also be mapped to projected climate conditions to estimate future range distributions.

Opportunity 2: Changes in plant phenology

Phenology, the study of the timing of events in biological life cycles, influences vegetation climate feedbacks by impacting both physical (i.e. sensible and latent heat fluxes) and biogeochemical (i.e. carbon uptake) processes (Peñuelas et al., 2009). As temperatures warm, many large geographical areas will experience longer growing seasons, and thus a greater proportion of the year with active vegetation. More growing vegetation impacts surface roughness, turbulent heat fluxes and canopy conductance (Bonan, 2008; Richardson et al., 2013), altering the surface energy balance in ways that lead to cooling (Su et al., 2020). Alternatively, early onset of springtime and leaf emergence could deplete soil moisture and lead to summer water stress, which would result in an increase in T_{surf} (Stéfanon et al., 2012; Wolf et al., 2016). Existing long-term (20+ years) data records of satellite T_{surf} offer opportunities to evaluate the impacts of changing phenology on vegetation climate feedbacks.

Temperatures experienced by plants (plant microclimates) both drive and respond to changes in phenology (Richardson et al., 2013; Vitasse et al., 2021). Although plant microclimate can vary significantly from T_{air} and change considerably over short spatial and temporal scales, broad-scale air and T_{surf} estimates are used more often than microclimate estimates in plant phenology (Zellweger et al., 2019). Considering the temporal resolution of these metrics, this can be problematic because temperature differentially influences plant process depending on phenology. For example, an extreme cold event during bud-break will have more of an impact than during the dormant season of a plant. A single metric calculated across an entire year (e.g. mean annual temperature) gives equal consideration to the thermal regime in both the dormant and growing seasons and still fails to reflect the temperature extremes that impact key stages of phenology (Körner & Hiltbrunner, 2018). Typically, such estimates occur at coarse spatial resolutions and fail to accurately reflect the actual temperatures experienced by plants. Characterizing plant microclimates using thermal remote sensing could better inform understanding of plant phenology, including impacts of shifting phenology on ecosystem structure and function.

Opportunity 3: Space-for-time

'Space-for-time' substitutions are used in ecology to predict ecological responses to changing climates. Such substitutions often incorporate sites spanning an environmental gradient in space, using changes in response across the gradient to project temporal trajectories (Johnson & Miyanishi, 2008; Pickett, 1989). For example, Klesse et al. (2020) leveraged the wide current environmental distribution of Douglas-fir to predict future distribution of the species based on the idea that the response of the species to a warmer future will follow the response of the individuals that currently occupy warmer locations. Remote sensing is ideally suited to inform space-for-time substitutions because satellite observations of the land surface provide wall-to-wall regional and global spatial coverage. The insight that thermal remote sensing can provide into photosynthetic acclimation and range shifts could be incorporated into space-for-time substitutions to make predictions about future species distributions, richness and turnover. For example, we could acquire thermal data at multiple latitudes or elevations, evaluate current metrics of plant stress or productivity and predict plant performance under future climate projections.

3 | USING THERMAL DATA

With advances in sensors, platforms and data availability, ecologists have increasing opportunities for the incorporation of thermal remote sensing into their own research, providing novel insight into the pressing questions in plant ecology from leaf to global scales. We will provide an overview of how ecologists can use thermal remote sensing data including a summary of TIR sensors and platforms, how the incorporation of additional remote sensing datasets can overcome some limitations inherent to TIR data, and provide a summary of some of the tools and resources available to aid in TIR data analysis.

Many remote sensing platforms and sensors, ranging from handheld devices to satellites, measure thermal data in various spatial, temporal and spectral resolutions (Figure 2). Instead of discussing individual current remote sensing platforms and sensors in detail, we highlight general characteristics of sensors in relation to the ecological scales addressed in this review. For interested readers, a summary of common airborne and satellite-based thermal remote sensing platforms is provided in Appendix S2 of Still, Powell, et al. (2019), and a comprehensive review of current and upcoming spaceborne instruments relevant to ecological applications are comprehensively reviewed in Ustin and Middleton (2021).

Handheld TIR cameras generate detailed images that can resolve fine-scale temperature variations within individual leaves. Thermal

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cameras designed for UAVs (uncooled microbolometers) can observe canopies to individual leaves, but require careful planning of sampling and subsequent data processing to collect accurate temperature data (see Kelly et al., 2019 and recommendations therein). Depending on the measurement height and field of view, TIR sensors mounted on aircraft or towers have the capability to observe both canopies and leaves.

Geostationary satellites rotate with Earth's orbit, providing high temporal resolutions (i.e. 15 min) for their field of view that allow investigation of diurnal patterns in LST beyond the canopy scale (~2 km; Xiao et al., 2021). Polar-orbiting satellites are typically in sun-synchronous orbit, with each overpass occurring at the same local time. Moderate spatial resolution (~120 m to 1 km in thermal bands), temporal resolutions ranging from every 16 days to twice per day, and long-term observation records facilitate monitoring of short- and long-term trends in T_{surf} beyond canopy scales. Instruments at low-inclination orbit, including those aboard the International Space Station (i.e. ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station, ECOSTRESS), have canopy-scale spatial resolution (70 m), with temporal coverage every 1–5 days occurring at different times of day, though lack polar coverage (Fisher et al., 2020). Thermal sensors are also being deployed on commercial spaceborne smallsats, which improve spatiotemporal coverage (Fisher et al., 2021; Lalli & Soenen, 2021).

Combining thermal remote sensing data with other types of remotely sensed data can greatly increase the information obtained, relative to analysis using a single sensor (Shiklomanov et al., 2019). This type of data fusion would allow for the consideration of additional ecologically relevant measurements (Stavros et al., 2017), help reduce uncertainty associated with TIR data products and expand the utility of thermal data across scales and ecosystem types. For example, fusion of multispectral and thermal UAV data provided better estimates of nitrogen concentration, chlorophyll *a* content and biomass than either data source alone (Maimaitijiang et al., 2017). In global change experiments, thermal remote sensing could provide frequent temporal measurements of high spatial resolution T_{surf} that could help characterize microclimates or pinpoint drivers of plant functioning and distribution (Fisher et al., 2021). Although repeat measurements at fine scale are ideal for many ecological applications, this has not been the case with TIR sensors mounted on satellites. This is especially problematic in systems with patchy vegetation coverage where a single pixel contains emitted TIR data not only from the plant(s) of interests, but also the background soil and senescent or dead/decomposing vegetation. However, this is rapidly evolving with upcoming launches of a constellation of smallsat TIR instruments to provide high spatial and high temporal resolution TIR measurements (Fisher et al., 2021). Vegetation information from other remote sensing sensors (e.g. indices calculated from visible and NIR reflectance, or structural information derived from LiDAR) can help constrain such analysis to only include areas with high vegetation coverage, resulting in TIR estimates that accurately represent plant temperature.

The utility of data from current and upcoming missions is dependent on free and open data access. US agencies and the ESA have adopted full, free and open data policies that have facilitated the widespread and productive use of satellite data by the ecological community. Significant advances over the past decade have made accessing and analysing remote sensing data easier than ever before. Data from satellite and airborne sensors are commonly divided into scenes or tiles that cover a specific amount of space on the Earth's surface. In ecological analyses, study areas can cover many tiles, causing data download sizes to increase exponentially over time. Alternatively, study areas can be smaller than a tile, causing scaling problems when TIR measurements are coarser than the actual area of interest. Subsetting tools, like NASA's Application for Extracting and Exploring Analysis-Ready Samples (AppEEARs) greatly reduces the memory and computational requirements previously required for remote sensing analyses by automatically subsetting and extracting geospatial data. Data can be extracted for specific point locations or as area subsets and downloaded as .csv or GeoTIFFs that can be readily imported into Geographic Information System (GIS) software with user-friendly point-and-click graphical user interfaces.

In terms of remote sensing data processing, Google Earth Engine (GEE) is a geospatial processing service and cloud computing platform that eliminates the need to download and analyse large satellite datasets on local computers. Amazon Web Services are also providing cloud-enabled access and processing to ever-increasing volumes of data, especially as spatiotemporal resolutions increase. Both the Python and R (R Core Team, 2020) programming languages have useful software packages for various aspects of thermal data processing including energy balance modelling (R 'tealeaves' (Okajima et al., 2012); Python 'pyTSEB' (Neito et al., 2021)), automated image processing workflows (Python 'PlantCV'; Gehan et al., 2017), thermal image correction (R 'ThermImage'; Tattersall, 2021), Python 'PyTirCam' (Calusi et al., 2020) and quantifying thermal heterogeneity (R 'Thermstats'; Senior et al., 2019). For those who would like to analyse data without coding, data are accessible as GeoTIFFs that can be readily imported into Geographic Information System (GIS) software such as QGIS with user-friendly point-and-click graphical user interfaces. The increase in data quality, quantity and access, alongside advances in analytics and visualization, presents many exciting opportunities to explore cutting-edge questions in plant ecology with thermal remote sensing data.

4 | CONCLUSIONS

Our motivations for this review were to present thermal remote sensing as a useful and tractable tool that can help resolve key ecological knowledge gaps. Remotely sensed TIR measurements data are coupled to plant physiology, ecology and biogeography, from leaf to global scales. At the leaf scale, thermal data can help us evaluate stomatal behaviour in response to varying environmental conditions. At the canopy scale, temperature estimates have fundamentally changed management of agricultural systems and presents similar opportunities in natural systems that have not yet been fully explored. Across broader scales, thermal

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data can provide insight into changing vegetation dynamics (e.g. range shifts, phenology) under future climate projections. Scaling challenges related to variation in leaf structural traits, canopy architecture and the background environment are important considerations, particularly at the canopy scale and in environments with sparse vegetation. Increases in data quantity and quality in combination with comprehensive open-source analytical tools offers opportunities for both novices and experts in remote sensing to conduct research evaluating open questions in plant ecology using remotely sensed thermal imagery. The potential of thermal remote sensing will be most fully realized when analysed in combination with other data sources, models and manipulative experiments. We hope these ideas help build a bridge between thermal remote sensing and plant ecology by providing new perspectives to leading-edge questions in plant biology.

AUTHORS' CONTRIBUTIONS

M.L.B. and M.M.F. developed the ideas and prepared the Figures; M.L.B. and M.M.F. drafted and coordinated the manuscript preparation; J.B.F. assisted with manuscript preparation and figure design. W.J. and K.B.K. reviewed the background literature. All authors contributed critically to the drafts and gave final approval for publication.

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CONFLICT OF INTEREST

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